1 The Succinctness Gap in Quantum Automata in a 1-Qubit Matrix Transformation Framework

Succinctness Gap in Quantum Automata

BS-CS-27-130

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The goal of this project is to solve the succinctness problem in quantum formal language theory by investigating the 1-Qubit Quantum Finite Automaton (QFA) through the lens of linear algebraic transformations. While QFAs are known to require exponentially fewer states than classical deterministic automata for certain languages, finding the exact bounds of this succinctness gap remains a critical open challenge. By conceptualizing quantum operations as continuous state transitions driven by 2×2 unitary matrices, we establish a rigorous mathematical framework to analyze quantum state complexity. We demonstrate the step-by-step processing of binary strings to calculate probability amplitudes, utilizing this matrix-driven approach as a foundational step toward systematically characterizing the exponential memory efficiency of quantum automata.

Keywords: Quantum Finite Automata, Formal Language Theory, Succinctness Gap, Unitary Matrices, State Complexity.

2 Trading Eye: Game Theory Method for Optimizing Stock-Market Trading Strategies

Trading Eye: A Game Theory Method for Building Trading Strategies in the Live Stock Market.

BS-CS-27-131

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The goal of this project is to construct *Trading Eye*: an algorithmic trading platform tailored to design, backtest, and optimize real-world financial strategies. It scans financial markets for the first ten or more assets—stocks or currencies—that meet a defined long-entry signal. The user is free to build customized strategies based on the basic indicators including Simple, Exponential, and Hull Moving Averages (SMAs, EMAs, HMAs), Volatility, Bollinger bands, and more. Trading Eye should also accommodate delta-neutral portfolios based on the Black-Scholes equation. The app simulates and visualizes profit margins in real time or using historical data. It supports multiple strategies,

each based on probabilistic models trained on past market behavior, and allocates investments using weighted distributions. Users can compare the performance of arbitrary strategies via clear, interactive graphs. The system is designed to aid traders in identifying high-probability opportunities and optimizing portfolio decisions with data-driven insights.

Keywords: algorithmic trading, real-time graphs, historical data, probability-based strategy, weighted investments, profit comparison.

3 Building a Turing Machine that Computes the Genetic Translation of a DNA Sequence and Outputs a Protein

Building a Turing Machine that Computes the Genetic Translation of a DNA Sequence and Outputs a Protein

BS-CS-27-132

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This project implements a Turing machine that computes a biological function: translating a DNA sequence over the alphabet $\{A, T, C, G\}$ into a corresponding protein or phenotypic characteristic. The input is a gene encoded as a finite DNA string, and the output is either a named protein (e.g., insulin, haemoglobin) or a trait determined by that gene. The alphabet $\{A, T, C, G\}$ represents the four nucleotide bases—adenine, thymine, cytosine, and guanine—that form the base pairs of the DNA double helix. The tape of a Turing machine closely resembles a single DNA strand, while a two-tape Turing machine mirrors the structure of the DNA double helix. The machine simulates transcription and translation by mapping codons to amino acids and applying rules of gene expression.

Keywords: Turing machine, DNA, protein synthesis, gene expression, codon mapping, molecular biology, computational biology, transcription, translation, DNA double helix, nucleotide bases, theoretical computer science

4 מחשוב קוונטי Quantum computing methods of improving time and space-complexity

מחשוב קוונטי

BS-CS-27-133

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The goal of this project is to check if the time and space complexity of computationally hard problems is genuinely improved by a quantum computer compared to classical systems. The project will specifically focus on evaluating a list of problems including the Traveling Salesperson Problem (TSP), Max-Cut, and Circuit-SAT. Quantum computers utilize qubits, which can exist in multiple states simultaneously due to the phenomena of superposition and entanglement. By exploring these parallel processing capabilities, we will design and implement quantum optimization algorithms (such as QAOA) to rigorously measure algorithmic performance and demonstrate potential practical quantum advantages in both time and space complexity.

Keywords: Quantum Computing, Time Complexity, Space Complexity, Traveling Salesperson Problem, Max-Cut, Circuit-SAT, QAOA.

5 שזירת קוונטית (QKD) Quantum Entanglement methods for mapping key distributions

שזירת קוונטית (QKD)

BS-CS-27-135

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The goal of this project is to model and simulate quantum entanglement to develop highly secure communication protocols focusing on quantum key-mapping and key distribution (QKD). Entanglement is a purely quantum phenomenon where the states of two or more particles become fundamentally correlated, enabling the secure generation and mapping of cryptographic keys across vast distances. This non-local physical property acts as the absolute cornerstone for advanced quantum key-mapping algorithms, ensuring that any external eavesdropping immediately disrupts the quantum state and is detected.

Keywords: Quantum Entanglement, Quantum Key-Mapping, Non-locality, Quantum Key Distribution, Cryptography.

6 אלגוריתם של שור לפענוח של RSA ללא מפתח ע"י פענוח קוונטי Shor's algorithm for decrypting RSA ciphertext in the absence of keys

אלגוריתם של שור לפענוח של RSA ללא מפתח ע"י פענוח קוונטי

BS-CS-27-136

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The goal of this project is to implement and analyze Shor's algorithm for factoring large integers, demonstrating the theoretical vulnerability of the RSA cryptosystem to quantum computing attacks. The security of RSA historically relies on the computational difficulty of prime factorization for classical computers. Shor's algorithm, however, cleverly leverages quantum Fourier transforms and periodicity finding to solve integer factorization in polynomial time, posing a critical threat to modern cryptographic paradigms.

Keywords: Shor's Algorithm, Prime Factorization, RSA Cryptography, Quantum Fourier Transform, Post-Quantum Cryptography.

7 אסטרטגיות מנצחות בשוק ההון, בלק שולס והיפוכים מקריים בשוק Stock market strategies based on the Black-Scholls equation and market reversal

אסטרטגיות מנצחות בשוק ההון, בלק שולס והיפוכים מקריים בשוק

BS-CS-27-137

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The goal of this project is to design robust algorithmic trading strategies by integrating the Black-Scholes model with statistical methods for detecting random market reversals. The Black-Scholes mathematical model estimates the theoretical value of put and call options based on dynamic variables like volatility and time to expiration. Combining this continuous-time finance model with stochastic calculus allows us to model random walks and construct delta neutral trading portfolios, successfully identifying probabilistic mean-reversion anomalies in asset prices for optimized trading.

Keywords: Black-Scholes Model, Algorithmic Trading, Mean Reversion, Stochastic Calculus, Put and Call Options, Delta Neutral Trading.

8 חישוב קומבינציות של גנים כדי לחשב הסתברות למחלה Combinatorial calculation of the probability of gene mutation

חישוב קומבינציות של גנים כדי לחשב הסתברות למחלה

BS-CS-27-138

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The goal of this project is to develop a computational model that calculates the probability of hereditary diseases based on complex gene combinations and specific mutations. Genetic predispositions are often polygenic, meaning multiple genes interact to influence a single phenotype or overall disease risk. By applying probabilistic models and combinatorial algorithms to genomic datasets, we can systematically map allele frequencies and interactions to accurately predict the statistical likelihood of specific genetic disorders.

Keywords: Computational Genomics, Polygenic Risk Score, Combinatorics, Probability Models, Bioinformatics.

9 אלגוריתם לפתרון משוואת דיפרנציאלית נומרית An numerical algorith for partial differentiatial systems with an complexity improvement of one order of time.

אלגוריתם לפתרון משוואת דיפרנציאלית נומרית

BS-CS-27-139

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The goal of this project is to implement and optimize numerical algorithms for solving complex ordinary and partial differential equations while rigorously calculating their time and space complexity. Many physical and biological systems are modeled using continuous differential equations lacking closed-form solutions. By focusing on the Runge-Kutta method and other discretization techniques,

we aim to approximate the dynamic behavior of these systems with high algorithmic precision. The project will specifically analyze the improved efficiency of optimized Runge-Kutta variations and benchmark their memory usage and execution time.

Keywords: Numerical Analysis, Differential Equations, Runge-Kutta, Discretization, Time Complexity, Space Complexity, Improved Efficiency.

10 בניית *LLM* - המודל המתמטי של *AI* Building a Large Language Model to solve analytical equations.

בניית *LLM* - המודל המתמטי של *AI*

BS-CS-27-140

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The goal of this project is to construct a scaled-down Large Language Model (LLM) to rigorously explore the linear algebra and deep mathematical foundations of modern artificial intelligence. By building neural networks from the ground up, the project will investigate non-linear transformations driven by the sigmoid model and other continuous activation functions. The architecture's underlying mathematics relies heavily on high-dimensional linear algebra, specifically focusing on complex matrix multiplications, matrix inversions for weight space analysis, and gradient descent optimization during backpropagation. This mathematical approach illuminates how dense vector embeddings and probabilistic token predictions fundamentally emerge from pure algebraic operations.

Keywords: Large Language Models, Linear Algebra, Matrix Inversions, Sigmoid Model, Deep Learning, Mathematical Optimization.